

Mark Scheme

Q1.

Question	Scheme	Marks	AOs
(a)(i)	$\frac{dy}{dx} = 20x^3 - 72x^2 + 84x - 32$	M1 A1	1.1b 1.1b
(ii)	$\frac{d^2y}{dx^2} = 60x^2 - 144x + 84$	A1ft	1.1b
		(3)	

(b)(i)	$x = 1 \Rightarrow \frac{dy}{dx} = 20 - 72 + 84 - 32$	M1	1.1b
	$\frac{dy}{dx} = 0$ so there is a stationary point at $x = 1$	A1	2.1
	Alternative for (b)(i)		
	$20x^3 - 72x^2 + 84x - 32 = 4(x-1)^2(5x-8) = 0 \Rightarrow x = \dots$	M1	1.1b
	When $x = 1$, $\frac{dy}{dx} = 0$ so there is a stationary point	A1	2.1
(b)(ii)	Note that in (b)(ii) there are no marks for <u>just</u> evaluating $\left(\frac{d^2y}{dx^2}\right)_{x=1}$		
	E.g. $\left(\frac{d^2y}{dx^2}\right)_{x=0.8} = \dots \left(\frac{d^2y}{dx^2}\right)_{x=1.2} = \dots$	M1	2.1
	$\left(\frac{d^2y}{dx^2}\right)_{x=0.8} > 0, \left(\frac{d^2y}{dx^2}\right)_{x=1.2} < 0$ Hence point of inflection	A1	2.2a
		(4)	
	Alternative 1 for (b)(ii)		
	$\left(\frac{d^2y}{dx^2}\right)_{x=1} = 60x^2 - 144x + 84 = 0$ (is inconclusive) $\left(\frac{d^3y}{dx^3}\right) = 120x - 144 \Rightarrow \left(\frac{d^3y}{dx^3}\right)_{x=1} = \dots$	M1	2.1
	$\left(\frac{d^2y}{dx^2}\right)_{x=1} = 0$ and $\left(\frac{d^3y}{dx^3}\right)_{x=1} \neq 0$ Hence point of inflection	A1	2.2a
	Alternative 2 for (b)(ii)		
	E.g. $\left(\frac{dy}{dx}\right)_{x=0.8} = \dots \left(\frac{dy}{dx}\right)_{x=1.2} = \dots$	M1	2.1
	$\left(\frac{dy}{dx}\right)_{x=0.8} < 0, \left(\frac{dy}{dx}\right)_{x=1.2} < 0$ Hence point of inflection	A1	2.2a
(7 marks)			

(a)(i)

M1: $x^n \rightarrow x^{n-1}$ for at least one power of x

A1: $\frac{dy}{dx} = 20x^3 - 72x^2 + 84x - 32$

(a)(ii)

A1ft: Achieves a correct $\frac{d^2y}{dx^2}$ for their $\frac{dy}{dx} = 20x^3 - 72x^2 + 84x - 32$

(b)(i)

M1: Substitutes $x = 1$ into their $\frac{dy}{dx}$

A1: Obtains $\frac{dy}{dx} = 0$ following a correct derivative and makes a conclusion which can be minimal

e.g. tick, QED etc. which may be in a preamble e.g. stationary point when $\frac{dy}{dx} = 0$ and then

shows $\frac{dy}{dx} = 0$

Alternative:

M1: Attempts to solve $\frac{dy}{dx} = 0$ by factorisation. This may be by using the factor of $(x - 1)$ or possibly using a calculator to find the roots and showing the factorisation. Note that they may divide by 4 before factorising which is acceptable. Need to either see either $4(x - 1)^2(5x - 8)$ or $(x - 1)^2(5x - 8)$ for the factorisation or $x = \frac{8}{5}$ and $x = 1$ seen as the roots.

A1: Obtains $x = 1$ and makes a conclusion as above

(b)(ii)

M1: Considers the value of the second derivative either side of $x = 1$. Do not be too concerned with the interval for the method mark.

(NB $\frac{d^2y}{dx^2} = (x-1)(60x-84)$ so may use this factorised form when considering $x < 1$, $x > 1$ for sign change of second derivative)

A1: Fully correct work including a correct $\frac{d^2y}{dx^2}$ with a reasoned conclusion indicating that the stationary point is a point of inflection. Sufficient reason is e.g. "sign change"/ " > 0 , < 0 ". If values are given they should be correct (but be generous with accuracy) but also just allow " > 0 " and " < 0 " provided they are correctly paired. The interval must be where $x < 1.4$

Alternative 1 for (b)(ii)

M1: Shows that second derivative at $x = 1$ is zero and then finds the third derivative at $x = 1$

A1: Fully correct work including a correct $\frac{d^2y}{dx^2}$ with a reasoned conclusion indicating that stationary point is a point of inflection. Sufficient reason is " $\neq 0$ " but must follow a correct third derivative and a correct value if evaluated. For reference $\left(\frac{d^3y}{dx^3}\right)_{x=1} = -24$

Alternative 2 for (b)(ii)

M1: Considers the value of the first derivative either side of $x = 1$. Do not be too concerned with the interval for the method mark.

A1: Fully correct work with a reasoned conclusion indicating that stationary point is a point of inflection. Sufficient reason is e.g. "same sign"/ "both negative"/ " < 0 , < 0 ". If values are given they should be correct (but be generous with accuracy). The interval must be where $x < 1.4$

x	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
$f(x)$	-32	-24.3	-17.92	-12.74	-8.64	-5.5	-3.2	-1.62	-0.64	-0.14	0
$f'(x)$	84	70.2	57.6	46.2	36	27	19.2	12.6	7.2	3	0

x	1.1	1.2	1.3	1.4	1.5	1.6	1.7
$f(x)$	-0.1	-0.32	-0.54	-0.64	-0.5	0	0.98
$f'(x)$	-1.8	-2.4	-1.8	0	3	7.2	12.6

Q2.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$6 = \pi r^2 h + \frac{2}{3} \pi r^3$	B1	This mark is given for a method to find the volume of the cylinder and the semi-hemisphere
	$A = 3\pi r^2 + 2\pi \left(\frac{6 - \frac{2}{3}\pi r^3}{\pi r} \right)$	M1 Q2.	This mark is given for a method to find the surface area of the tank
		A1	This mark is given for finding an expression for the surface area of the tank
	$A = 3\pi r^2 + \frac{12}{r} - \frac{4\pi r^2}{3} = \frac{12}{r} + \frac{5\pi r^2}{3}$	A1	This mark is given for a fully correct proof to show the surface area of the tank as required
(b)	$A = \frac{12}{r} + \frac{5\pi r^2}{3} \Rightarrow \frac{dA}{dr} = -\frac{12}{r^2} + \frac{10\pi r}{3}$	M1	This mark is given for a method to differentiate to find r
		A1	This mark is given for accurately differentiating to find r
	When $\frac{dA}{dr} = 0$, $-\frac{12}{r^2} + \frac{10\pi r}{3} = 0$ $r^3 = \frac{18}{5\pi}$	M1	This mark is given for a method to set $\frac{dA}{dr} = 0$ to find a value for r
	$r = 1.046$	A1	This mark is given for finding the radius for which the surface area is a minimum
(c)	$A = \frac{12}{1.046} + \frac{5\pi(1.046)^2}{3}$	M1	This mark is given for a method to substitute a value for r
	$A = 17 \text{ m}^2$	A1	This mark is given for correctly finding the minimum surface area of the tank (to the nearest integer)
(Total 10 marks)			

Question	Scheme	Marks	AOs
(a)	$k = e^2$ or $x \neq e^2$	B1	2.2a
		(1)	
(b)	$g'(x) = \frac{(\ln x - 2) \times \frac{3}{x} - (3 \ln x - 7) \times \frac{1}{x}}{(\ln x - 2)^2} = \frac{1}{x(\ln x - 2)^2}$ or $g'(x) = \frac{d}{dx} \left(3 - (\ln(x) - 2)^{-1} \right) = (\ln x - 2)^{-2} \times \frac{1}{x} = \frac{1}{x(\ln x - 2)^2}$ or $g'(x) = (\ln x - 2)^{-1} \times \frac{3}{x} - (3 \ln x - 7)(\ln x - 2)^{-2} \times \frac{1}{x} = \frac{1}{x(\ln x - 2)^2}$	M1 A1	1.1b 2.1
	As $x > 0$ (or $1/x > 0$) AND $\ln x - 2$ is squared so $g'(x) > 0$	A1cso	2.4
		(3)	
(c)	Attempts to solve either $3 \ln x - 7 \dots 0$ or $\ln x - 2 \dots 0$ or $3 \ln a - 7 \dots 0$ or $\ln a - 2 \dots 0$ where $\dots$ is " $=$ " or " $>$ " to reach a value for x or a but may be seen as an inequality e.g. $x > \dots$ or $a > \dots$	M1	3.1a
	$0 < a < e^2$, $a > e^{\frac{7}{3}}$	A1	2.2a
		(2)	
(6 marks)			

Notes:

(a)

B1: Deduces $k = e^2$ or $x \neq e^2$ Condone $k = \text{awrt } 7.39$ or $x \neq \text{awrt } 7.39$

(b)

M1: Attempts to differentiate via the quotient rule and with $\ln x \rightarrow \frac{1}{x}$ so allow for:

$$\frac{d}{dx}(g(x)) = \frac{(\ln x - 2) \times \frac{\alpha}{x} - (3 \ln x - 7) \times \frac{\beta}{x}}{(\ln x - 2)^2}, \quad \beta > 0$$

But a correct rule may be implied by their u, v, u', v' followed by applying $\frac{vu' - uv'}{v^2}$ etc.Alternatively attempts to write $g(x) = \frac{3 \ln(x) - 7}{\ln(x) - 2} = 3 - (\ln(x) - 2)^{-1}$ and attempts the chain rule so allow for:

$$3 - (\ln(x) - 2)^{-1} \rightarrow (\ln(x) - 2)^{-2} \times \frac{\alpha}{x}$$

Alternatively writes $g(x) = (3 \ln(x) - 7)(\ln(x) - 2)^{-1}$ and attempts the product rule so allow for:

$$g'(x) = (\ln x - 2)^{-1} \times \frac{\alpha}{x} - (3 \ln x - 7)(\ln x - 2)^{-2} \times \frac{\beta}{x}$$

In general condone missing brackets for the M mark. E.g. if they quote $u = 3 \ln v - 7$ and $v = \ln x - 2$ and don't make the differentiation easier, they can be awarded this mark for applying the correct rule. Also allow this mark if they quote the correct quotient rule but only have v rather than v^2 in the denominator.

A1: $\frac{1}{x(\ln x - 2)^2}$ Allow $\frac{\frac{1}{x}}{(\ln x - 2)^2}$ i.e. we need to see the numerator simplified to $1/x$

Note that some candidates establish the correct numerator and correct denominator independently and provided they obtain the correct expressions, this mark can be awarded.

But allow a correctly expanded denominator.

A1cso: States that as $x > 0$ AND $\ln x - 2$ is squared so $g'(x) > 0$

(c)

M1: Attempts to solve either $3 \ln x - 7 = 0$ or $\ln x - 2 = 0$ or using inequalities e.g. $3 \ln x - 7 > 0$ A1: $0 < a < e^2$, $a > e^{\frac{7}{3}}$

Q4.

Question	Scheme	Marks	AOs
(a)	$\ln x \rightarrow \frac{1}{x}$	B1	1.1a
	Method to differentiate $\frac{4x^2 + x}{2\sqrt{x}}$ – see notes	M1	1.1b
	E.g. $2 \times \frac{3}{2}x^{\frac{1}{2}} + \frac{1}{2} \times \frac{1}{2}x^{-\frac{1}{2}}$	A1	1.1b
	$\frac{dy}{dx} = 3\sqrt{x} + \frac{1}{4\sqrt{x}} - \frac{4}{x} = \frac{12x^2 + x - 16\sqrt{x}}{4x\sqrt{x}}$ *	A1*	2.1
	(4)		
(b)	$12x^2 + x - 16\sqrt{x} = 0 \Rightarrow 12x^{\frac{3}{2}} + x^{\frac{3}{2}} - 16 = 0$	M1	1.1b
	E.g. $12x^{\frac{3}{2}} = 16 - \sqrt{x}$	dM1	1.1b
	$x^{\frac{3}{2}} = \frac{4}{3} - \frac{\sqrt{x}}{12} \Rightarrow x = \left(\frac{4}{3} - \frac{\sqrt{x}}{12}\right)^{\frac{2}{3}}$ *	A1*	2.1
	(3)		
(c)	$x_2 = \sqrt[3]{\left(\frac{4}{3} - \frac{\sqrt{2}}{12}\right)^2}$	M1	1.1b
	$x_2 = \text{awrt } 1.13894$	A1	1.1b
	$x = 1.15650$	A1	2.2a
	(3)		
(10 marks)			

Notes:

(a)

B1: Differentiates $\ln x \rightarrow \frac{1}{x}$ seen or implied

M1: Correct method to differentiate $\frac{4x^2 + x}{2\sqrt{x}}$:

Look for $\frac{4x^2 + x}{2\sqrt{x}} \rightarrow \dots x^{\frac{3}{2}} + \dots x^{\frac{1}{2}}$ being then differentiated to $Px^{\frac{1}{2}} + \dots$ or $\dots + Qx^{-\frac{1}{2}}$

Alternatively uses the quotient rule on $\frac{4x^2 + x}{2\sqrt{x}}$.

Condone slips but if rule is not quoted expect $\left(\frac{dy}{dx}\right) = \frac{2\sqrt{x}(Ax+B) - (4x^2+x)Cx^{\frac{1}{2}}}{(2\sqrt{x})^2} (A, B, C > 0)$

But a correct rule may be implied by their u, v, u', v' followed by applying $\frac{vu' - uv'}{v^2}$ etc.

Alternatively uses the product rule on $(4x^2 + x)(2\sqrt{x})^{-1}$

Condone slips but expect $\left(\frac{dy}{dx}\right) = Ax^{-\frac{1}{2}}(Bx+C) + D(4x^2+x)x^{-\frac{3}{2}} (A, B, C > 0)$

In general condone missing brackets for the M mark. If they quote $u = 4x^2 + x$ and $v = 2\sqrt{x}$ and don't make the differentiation easier, they can be awarded this mark for applying the correct rule. Also allow this mark if they quote the correct quotient rule but only have v rather than v^2 in the denominator.

A1: Correct differentiation of $\frac{4x^2 + x}{2\sqrt{x}}$ although may not be simplified.

Examples: $\left(\frac{dy}{dx}\right) = \frac{2\sqrt{x}(8x+1) - (4x^2+x)x^{-\frac{1}{2}}}{(2\sqrt{x})^2}, \frac{1}{2}x^{-\frac{1}{2}}(8x+1) - \frac{1}{4}(4x^2+x)x^{-\frac{3}{2}}, 2 \times \frac{3}{2}x^{\frac{1}{2}} + \frac{1}{2} \times \frac{1}{2}x^{-\frac{1}{2}}$

A1*: Obtains $\frac{dy}{dx} = \frac{12x^2 + x - 16\sqrt{x}}{4x\sqrt{x}}$ via $3\sqrt{x} + \frac{1}{4\sqrt{x}} - \frac{4}{x}$ or a correct application of the quotient or product rule

and with sufficient working shown to reach the printed answer.

There must be no errors e.g. missing brackets.

(b)

M1: Sets $12x^2 + x - 16\sqrt{x} = 0$ and divides by $\sqrt{x}$ or equivalent e.g. divides by x and multiplies by $\sqrt{x}$

dM1: Makes the term in $x^{\frac{3}{2}}$ the subject of the formula

A1*: A correct and rigorous argument leading to the given solution.

Alternative - working backwards:

$$x = \left(\frac{4}{3} - \frac{\sqrt{x}}{12}\right)^{\frac{2}{3}} \Rightarrow x^{\frac{3}{2}} = \frac{4}{3} - \frac{\sqrt{x}}{12} \Rightarrow 12x^{\frac{3}{2}} = 16 - \sqrt{x} \Rightarrow 12x^2 = 16\sqrt{x} - x \Rightarrow 12x^2 - 16\sqrt{x} + x = 0$$

M1: For raising to power of $3/2$ both sides. **dM1**: Multiplies through by $\sqrt{x}$. **A1**: Achieves printed answer and makes a minimal comment e.g. tick, #, QED, true etc.

(c)

M1: Attempts to use the iterative formula with $x_1 = 2$. This is implied by sight of $x_2 = \left(\frac{4}{3} - \frac{\sqrt{2}}{12}\right)^{\frac{2}{3}}$ or awrt 1.14

A1: $x_2 = \text{awrt } 1.13894$

A1: Deduces that $x = 1.15650$

Q5.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$y = x^x \Rightarrow \ln y = x \ln x$	M1	This mark is for a method to find the x -coordinate of the turning point of C by taking logarithms
	$\ln y = x \ln x \Rightarrow \frac{1}{y} \frac{dy}{dx} = \ln x + 1$	M1	This mark is given for a method using implicit differentiation
		A1	This mark is given for a correct expression for $\frac{1}{y} \frac{dy}{dx}$
	Setting $\frac{dy}{dx} = 0, \ln x + 1 = 0$	M1	This mark is given for a method for finding the turning point of C by setting $\frac{dy}{dx} = 0$
	$x = e^{-1}$	A1	This mark is given for correctly finding a value for the x -coordinate of the turning point of C

(b)	$1.5^{1.5} = 1.837\dots, 1.6^{1.6} = 2.121\dots$	M1	This mark is given for substituting 1.5 and 1.6 into $y = x^x$
	The curve C contains the points (1.5, 1.8) and (1.6, 2.1). At $P, y = 2$ Since C is continuous, $1.5 < \alpha < 1.6$	A1	This mark is given for a valid explanation that C contains the points (1.5, 1.8) and (1.6, 2.1) and is continuous
(c)	$x_1 = 1.5$ $x_2 = 2 \times 1.5^{-0.5} = 1.633$	M1	This mark is given for finding a correct value for x_2
	$x_3 = 2 \times 1.633^{-0.633} = 1.466$ $x_4 = 2 \times 1.466^{-0.466} = 1.673$	A1	This mark is given for finding a correct value for x_4
(d)	For example: x_n oscillates is periodic is non-convergent	B1	This mark is given for a valid statement about the long-term behaviour of x_n
	between 1 and 2	B1	This mark is given for stating that the behaviour is between 1 and 2
(Total 11 marks)			

Q6.

Question	Scheme	Marks	AOs
(a)	Attempts $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{4\sec^2 t \tan t}{2\sec^2 t} (= 2 \tan t)$	M1 A1	1.1b 1.1b
	At $t = \frac{\pi}{4}, \frac{dy}{dx} = 2, x = 3, y = 7$	M1	2.1
	Attempts equation of normal $y - 7 = -\frac{1}{2}(x - 3)$	M1	1.1b
	$y = -\frac{1}{2}x + \frac{17}{2} \quad *$	A1*	2.1
		(5)	
(b)	Attempts to use $\sec^2 t = 1 + \tan^2 t \Rightarrow \frac{y-3}{2} = 1 + \left(\frac{x-1}{2}\right)^2$	M1	3.1a
	$\Rightarrow y - 3 = 2 + \frac{(x-1)^2}{2} \Rightarrow y = \frac{1}{2}(x-1)^2 + 5 \quad *$	A1*	2.1
		(2)	

	(b) Alternative 1:		
	$y = \frac{1}{2}(x-1)^2 + 5 = \frac{1}{2}(2 \tan t + 1 - 1)^2 + 5$ $= \frac{1}{2} 4 \tan^2 t + 5 = 2(\sec^2 t - 1) + 5$	M1	3.1a
	$= 2 \sec^2 t + 3 = y^*$	A1	2.1
	(b) Alternative 2:		
	$x = 2 \tan t + 1 \Rightarrow t = \tan^{-1}\left(\frac{x-1}{2}\right) \Rightarrow y = 2 \sec^2\left(\tan^{-1}\left(\frac{x-1}{2}\right)\right) + 3$ $\Rightarrow y = 2\left(1 + \tan^2\left(\tan^{-1}\left(\frac{x-1}{2}\right)\right)\right) + 3$	M1	3.1a
	$\Rightarrow y = 2\left(1 + \left(\frac{x-1}{2}\right)^2\right) + 3 = \frac{1}{2}(x-1)^2 + 5^*$	A1	2.1
	(b) Alternative 3:		
	$\frac{dy}{dx} = 2 \tan t = x - 1 \Rightarrow y = \int (x-1) dx = \frac{x^2}{2} - x + c$ $(3, 7) \rightarrow 7 = \frac{3^2}{2} - 3 + c \Rightarrow c = \frac{11}{2}$	M1	3.1a
	$\frac{x^2}{2} - x + \frac{11}{2} = \frac{1}{2}(x^2 - 2x) + \frac{11}{2} = \frac{1}{2}(x-1)^2 - \frac{1}{2} + \frac{11}{2} = \frac{1}{2}(x-1)^2 + 5^*$	A1	2.1

(c)	Attempts the lower limit for k :		
	$\frac{1}{2}(x-1)^2 + 5 = -\frac{1}{2}x + k \Rightarrow x^2 - x + (11 - 2k) = 0$ $b^2 - 4ac = 1 - 4(11 - 2k) = 0 \Rightarrow k = \dots$	M1	2.1
	$(k =) \frac{43}{8}$	A1	1.1b
	Attempts the upper limit for k :		
	$(x, y)_{t=-\frac{\pi}{4}} : t = -\frac{\pi}{4} \Rightarrow x = 2 \tan\left(-\frac{\pi}{4}\right) + 1 = -1, y = 2 \sec^2\left(-\frac{\pi}{4}\right) + 3 = 7$ $(-1, 7), y = -\frac{1}{2}x + k \Rightarrow 7 = \frac{1}{2} + k \Rightarrow k = \dots$	M1	2.1
	$(k =) \frac{13}{2}$	A1	1.1b
	$\frac{43}{8} < k \leq \frac{13}{2}$	A1	2.2a
(5)			
(12 marks)			
Notes:			

(a) **Must use parametric differentiation to score any marks in this part and not e.g. Cartesian form**

M1: For the key step of attempting $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$. There must be some attempt to differentiate both

parameters however poor and divide the right way round so using $\frac{dy}{dx} = \frac{y}{x}$ scores M0.

This may be implied by e.g. $\frac{dx}{dt} = 2\sec^2 t$, $\frac{dy}{dt} = 4\sec^2 t \tan t$, $t = \frac{\pi}{4} \Rightarrow \frac{dx}{dt} = 4$, $\frac{dy}{dt} = 8 \Rightarrow \frac{dy}{dx} = 2$

A1: $\frac{dy}{dx} = \frac{4\sec^2 t \tan t}{2\sec^2 t}$. Correct expression in any form. May be implied as above.

Condone the confusion with variables as long as the intention is clear e.g.

$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{4\sec^2 x \tan x}{2\sec^2 x} (= 2 \tan x)$ and allow subsequent marks if this is interpreted correctly

M1: For attempting to find the values of x , y and the gradient at $t = \frac{\pi}{4}$ AND getting at least two correct.

Follow through on their $\frac{dy}{dx}$ so allow for any two of $x = 3$, $y = 7$, $\frac{dy}{dx} = 2$ (or their $\frac{dy}{dx}$ at $t = \frac{\pi}{4}$)

Note that the $x = 3$, $y = 7$ may be seen as e.g. (3, 7) on the diagram. There must be a non-trivial

$\frac{dy}{dx}$ for this mark e.g. they must have a $\frac{dy}{dx}$ to substitute into.

M1: For a correct attempt at the normal equation using their x and y at $t = \frac{\pi}{4}$ with the negative

reciprocal of their $\frac{dy}{dx}$ at $t = \frac{\pi}{4}$ having made some attempt at $\frac{dy}{dx}$ and all correctly placed.

For attempts using $y = mx + c$ they must reach as far as a value for c using their x and y at $t = \frac{\pi}{4}$

with the negative reciprocal of their $\frac{dy}{dx}$ at $t = \frac{\pi}{4}$ all correctly placed.

A1*: Proceeds with a clear argument to the given answer with no errors.

(b)

M1: Attempts to use $\sec^2 t = 1 + \tan^2 t$ oe to obtain an equation involving y and $(x-1)^2$

E.g. as above or e.g. $y = 2\sec^2 t + 3 = 2\left(1 + \tan^2 t\right) + 3 = 2\left(1 + \left(\frac{x-1}{2}\right)^2\right) + 3$ for M1 and then

$$y = \frac{1}{2}(x-1)^2 + 5^* \text{ for A1}$$

A1*: Proceeds with a clear argument to the given answer with no errors

Alternative 1:

M1: Uses the given result, substitutes for x and attempts to use $\sec^2 t = 1 + \tan^2 t$ oe

A1: Proceeds with a clear argument to the y parameter and makes a (minimal) conclusion e.g. “ $= y$ ”
QED, hence proven etc.

Alternative 2:

M1: Uses the x parameter to obtain t in terms of arctan, substitutes into y and attempts to use

$$\sec^2 t = 1 + \tan^2 t \text{ oe}$$

A1: Proceeds with a clear argument to the given answer with no errors

Alternative 3:

M1: Uses $\frac{dy}{dx}$ from part (a) to express $\frac{dy}{dx}$ in terms of x , integrates and uses $(3, 7)$ to find “ c ” to reach a Cartesian equation.

A1: Proceeds with a clear argument to the given answer with no errors

Allow the marks for (b) to score anywhere in their solution e.g. if they find the Cartesian equation in part (a)

(c)

M1: A full attempt to find the lower limit for k .

$$\frac{1}{2}(x-1)^2 + 5 = -\frac{1}{2}x + k \Rightarrow x^2 - x + (11 - 2k) = 0 \Rightarrow b^2 - 4ac = 1 - 4(11 - 2k) = 0 \Rightarrow k = \dots$$

Score **M1** for setting $\frac{1}{2}(x-1)^2 + 5 = -\frac{1}{2}x + k$, rearranging to 3TQ form and attempts $b^2 - 4ac \dots 0$

e.g. $b^2 - 4ac > 0$ or e.g. $b^2 - 4ac < 0$ correctly to find a value for k .

A1: $k = \frac{43}{8}$ oe. Look for this value e.g. may appear in an inequality e.g. $k > \frac{43}{8}$, $k < \frac{43}{8}$

An alternative method using calculus for lower limit:

$$y = \frac{1}{2}(x-1)^2 + 5 \Rightarrow \frac{dy}{dx} = x-1, x-1 = -\frac{1}{2} \Rightarrow x = \frac{1}{2}$$

$$x = \frac{1}{2} \Rightarrow y = \frac{1}{2}\left(\frac{1}{2}-1\right)^2 + 5 = \frac{41}{8}$$

$$y = -\frac{1}{2}x + k \Rightarrow \frac{41}{8} = -\frac{1}{4} + k \Rightarrow k = \dots$$

Score **M1** for $\frac{dy}{dx}$ = “a linear expression in x ”, sets $= -\frac{1}{2}$, solves a linear equation to find x and

then substitutes into the given result in (b) to find y and then uses $y = -\frac{1}{2}x + k$ to find a value

for k . **A1:** $k = \frac{43}{8}$ oe. Look for this value e.g. may appear in an inequality e.g. $k > \frac{43}{8}$, $k < \frac{43}{8}$

An alternative method using parameters for lower limit:

$$\begin{aligned}
 y &= -\frac{1}{2}x + k \Rightarrow 2\sec^2 t + 3 = -\frac{1}{2}(2\tan t + 1) + k \\
 \Rightarrow 2(1 + \tan^2 t) + 3 &= -\frac{1}{2}(2\tan t + 1) + k \Rightarrow 2\tan^2 t + \tan t + 5.5 - k = 0 \\
 b^2 - 4ac &= 0 \Rightarrow 1 - 4 \times 2(5.5 - k) = 0 \Rightarrow k = \frac{43}{8}
 \end{aligned}$$

Score M1 for substituting parametric form of x and y into $y = -\frac{1}{2}x + k$, uses $\sec^2 t = 1 + \tan^2 t$

rearranges to 3TQ form and attempts $b^2 - 4ac \dots 0$ or e.g. $b^2 - 4ac > 0$ or $b^2 - 4ac < 0$ correctly to find a value for k .

A1: $k = \frac{43}{8}$ oe. Look for this value e.g. may appear in an inequality e.g. $k > \frac{43}{8}$, $k < \frac{43}{8}$

M1: A full attempt to find the upper limit for k . This requires an attempt to find the value of x and the value of y using $t = -\frac{\pi}{4}$, the substitution of these values into $y = -\frac{1}{2}x + k$ and solves for k .

A1: $k = \frac{13}{2}$. Look for this value e.g. may appear in an inequality.

A1: Deduces the correct range for k : $\frac{43}{8} < k \leq \frac{13}{2}$

Allow equivalent notation e.g. $\left(k \leq \frac{13}{2} \text{ and } k > \frac{43}{8}\right)$, $\left(k \leq \frac{13}{2} \cap k > \frac{43}{8}\right)$, $\left(\frac{43}{8}, \frac{13}{2}\right]$

But not e.g. $\left(k \leq \frac{13}{2}, k > \frac{43}{8}\right)$, $\left(k \leq \frac{13}{2} \cup k > \frac{43}{8}\right)$, $\left(k \leq \frac{13}{2} \text{ or } k > \frac{43}{8}\right)$ and do not allow if in terms of x .

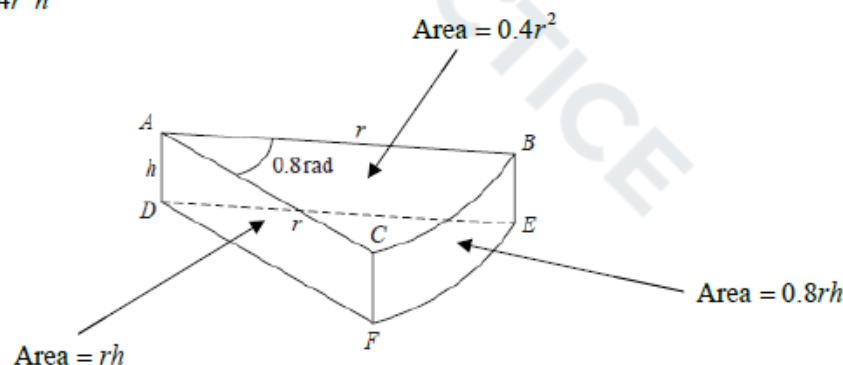
Allow equivalent exact values for $\frac{43}{8}$, $\frac{13}{2}$

There may be other methods for finding the upper limit which are valid. If you are in any doubt if a method deserves credit then use Review.

Q7.

Question	Scheme	Marks	AOs
(a)	Sets up an allowable equation using volume = 240 E.g. $\frac{1}{2}r^2 \times 0.8h = 240 \Rightarrow h = \frac{600}{r^2}$ o.e.	M1 A1	3.4 1.1b
	Attempts to substitute their $h = \frac{600}{r^2}$ into $(S =) \frac{1}{2}r^2 \times 0.8 + \frac{1}{2}r^2 \times 0.8 + 2rh + 0.8rh$	dM1	3.4
	$S = 0.8r^2 + 2.8rh = 0.8r^2 + 2.8 \times \frac{600}{r} = 0.8r^2 + \frac{1680}{r}$ *	A1*	2.1
	(4)		
(b)	$\left(\frac{dS}{dr}\right) = 1.6r - \frac{1680}{r^2}$	M1 A1	3.1a 1.1b
	Sets $\frac{dS}{dr} = 0 \Rightarrow r^3 = 1050$ $r = \text{awrt } 10.2$	dM1 A1	2.1 1.1b
	(4)		
(c)	Attempts to substitute their positive r into $\left(\frac{d^2S}{dr^2}\right) = 1.6 + \frac{3360}{r^3}$ and considers its value or sign	M1	1.1b
	E.g. Correct $\frac{d^2S}{dr^2} = 1.6 + \frac{3360}{r^3}$ with $\frac{d^2S}{dr^2} \bigg _{r=10.2} = 5 > 0$ proving a minimum value of S	A1	1.1b
		(2)	
(10 marks)			
Notes:			

$$\text{Volume} = 0.4r^2h$$



$$\text{Total surface area} = 2rh + 0.8r^2 + 0.8rh$$



(a)

M1: Attempts to use the fact that the volume of the toy is 240 cm^3 Sight of $\frac{1}{2}r^2 \times 0.8 \times h = 240$ leading to $h = \dots$ or $rh = \dots$ scores this markBut condone an equation of the correct form so allow for $kr^2h = 240 \Rightarrow h = \dots$ or $rh = \dots$ **A1:** A correct expression for $h = \frac{600}{r^2}$ or $rh = \frac{600}{r}$ which may be left unsimplified.This may be implied when you see an expression for S or part of S E.g. $2rh = 2r \times \frac{600}{r^2}$ **dM1:** Attempts to substitute their $h = \frac{a}{r^2}$ o.e. such as $hr = \frac{a}{r}$ into a correct expression for S Sight of $\frac{1}{2}r^2 \times 0.8 + \frac{1}{2}r^2 \times 0.8 + rh + rh + 0.8rh$ with an appropriate substitutionSimplified versions such as $0.8r^2 + 2rh + 0.8rh$ used with an appropriate substitution is fine.**A1*:** Correct work leading to the given result. $S =$, $SA =$ or surface area = must be seen at least once in the correct place

The method must be made clear so expect to see evidence. For example

$$S = 0.8r^2 + 2rh + 0.8rh \Rightarrow S = 0.8r^2 + 2r \times \frac{600}{r^2} + 0.8r \times \frac{600}{r^2} \Rightarrow S = 0.8r^2 + \frac{1680}{r} \text{ would be fine.}$$

(b) There is no requirement to see $\frac{dS}{dr}$ in part (b). It may even be called $\frac{dy}{dx}$.**M1:** Achieves a derivative of the form $pr \pm \frac{q}{r^2}$ where p and q are non-zero constants**A1:** Achieves $\left(\frac{dS}{dr}\right) = 1.6r - \frac{1680}{r^2}$ **dM1:** Sets or implies that their $\frac{dS}{dr} = 0$ and proceeds to $mr^3 = n$, $m \times n > 0$. It is dependent upon acorrect attempt at differentiation. This mark may be implied by a correct answer to their $pr - \frac{q}{r^2} = 0$ **A1:** $r = \text{awrt } 10.2$ or $\sqrt[3]{1050}$

(c)

M1: Attempts to substitute their positive r (found in (b)) into $\left(\frac{d^2S}{dr^2} = \right) e \pm \frac{f}{r^3}$ where e and f are non zero and finds its value or sign.Alternatively considers the sign of $\left(\frac{d^2S}{dr^2} = \right) e \pm \frac{f}{r^3}$ (at their positive r found in (b))Condone the $\frac{d^2S}{dr^2}$ to be $\frac{d^2y}{dx^2}$ or being absent, but only for this mark.**A1:** States that $\frac{d^2S}{dr^2}$ or $S'' = 1.6 + \frac{3360}{r^3} = \text{awrt } 5 > 0$ proving a minimum value of S This is dependent upon having achieved $r = \text{awrt } 10$ and a correct $\frac{d^2S}{dr^2} = 1.6 + \frac{3360}{r^3}$ It can be argued without finding the value of $\frac{d^2S}{dr^2}$. E.g. $\frac{d^2S}{dr^2} = 1.6 + \frac{3360}{r^3} > 0$ as $r > 0$, so minimum value of S . For consistency it is also dependent upon having achieved $r = \text{awrt } 10$ Do NOT allow $\frac{d^2y}{dx^2}$ for this mark

Q8.

Question	Scheme	Marks	AOs
(a)	$f(x) = (x-2)^2 \pm \dots$	M1	1.2
	$f(x) = (x-2)^2 + 1$	A1	1.1b
		(2)	
(b)(i)	$P = (0, 5)$	B1	1.1b
(b)(ii)	$Q = (2, 1)$	B1ft	1.1b
		(2)	
(4 marks)			
Notes			

(a)

M1: Achieves $(x-2)^2 \pm \dots$ or states $a = -2$

A1: Correct expression $(x-2)^2 + 1$ ISW after sight of this

Condone $a = -2$ and $b = 1$. Condone $(x-2)^2 + 1 = 0$

(b)

(i) B1: Correct coordinates for P . Allow to be expressed $x = 0, y = 5$

(ii) B1ft: Correct coordinates for Q . Allow to be expressed $x = 2, y = 1$ (Score for the correct answer or follow through their part (a) so allow $(-a, b)$ where a and b are numeric)

Score in any order if they state $P = (0, 5)$ and $Q = (2, 1)$

.....
Allow part (b) to be awarded from a sketch. So award

First B1 from a sketch crossing the y-axis at 5

Second B1 from a sketch with minimum at $(2, 1)$
.....

Q9.

Question	Scheme	Marks	AOs
(a)	$f'(x) = 2x + \frac{4x-4}{2x^2-4x+5}$	M1 A1	1.1b 1.1b
	$2x + \frac{4x-4}{2x^2-4x+5} = 0 \Rightarrow 2x(2x^2-4x+5) + 4x-4 = 0$	dM1	1.1b
	$2x^3 - 4x^2 + 7x - 2 = 0^*$	A1*	2.1
	(4)		
(b)	(i) $x_2 = \frac{1}{7}(2 + 4(0.3)^2 - 2(0.3)^3)$	M1	1.1b
	$x_2 = 0.3294$	A1	1.1b
	(ii) $x_4 = 0.3398$	A1	1.1b
	(3)		
(c)	$h(x) = 2x^3 - 4x^2 + 7x - 2$ $h(0.3415) = 0.00366... \quad h(0.3405) = -0.00130...$	M1	3.1a
	States: - there is a change of sign $f'(x)$ is continuous $\alpha = 0.341$ to 3dp	A1	2.4
	(2)		
	(9 marks)		
Notes			

(a)

M1: Differentiates $\ln(2x^2 - 4x + 5)$ to obtain $\frac{g(x)}{2x^2 - 4x + 5}$ where $g(x)$ could be 1

A1: For $f'(x) = 2x + \frac{4x-4}{2x^2-4x+5}$

dM1: Sets their $f'(x) = ax + \frac{g(x)}{2x^2-4x+5} = 0$ and uses "correct" algebra, condoning slips, to obtain a

cubic equation. E.g Look for $ax(2x^2 - 4x + 5) \pm g(x) = 0$ o.e. , condoning slips, followed by some attempt to simplify

A1*: Achieves $2x^3 - 4x^2 + 7x - 2 = 0$ with no errors. (The dM1 mark must have been awarded)

(b)(i)

M1: Attempts to use the iterative formula with $x_1 = 0.3$. If no method is shown award for $x_2 = \text{awrt } 0.33$

A1: $x_2 = \text{awrt } 0.3294$ Note that $\frac{1153}{3500}$ is correct

Condone an incorrect suffix if it is clear that a correct value has been found

(b)(ii)

A1: $x_4 = \text{awrt } 0.3398$ Condone an incorrect suffix if it is clear that a correct value has been found

(c)

M1: Attempts to substitute $x = 0.3415$ and $x = 0.3405$ into a suitable function and gets one value correct (rounded or truncated to 1 sf). It is allowable to use a tighter interval that contains the root 0.340762654

Examples of suitable functions are $2x^3 - 4x^2 + 7x - 2$, $x - \frac{1}{7}(4x^2 - 2x^3 + 2)$ and $f'(x)$ as this has been

found in part (a) with $f'(0.3405) = -0.00067...$, $f'(0.3415) = (+) 0.0018$

There must be sufficient evidence for the function, which would be for example, a statement such as $h(x) = 2x^3 - 4x^2 + 7x - 2$ or sight of embedded values that imply the function, not just a value or values

even if both are correct. Condone $h(x)$ being mislabelled as f

$h(0.3415) = 2 \times 0.3415^3 - 4 \times 0.3415^2 + 7 \times 0.3415 - 2$

A1: Requires

- both calculations correct (rounded or truncated to 1sf)
- a statement that there is a change in sign and that the function is continuous
- a minimal conclusion e.g. $\checkmark$ proven, $\alpha = 0.341$, root

Q10.

Question	Scheme	Marks	AOs
(a)	$f(x) = 4(x^2 - 2)e^{-2x}$		
	Differentiates to $e^{-2x} \times 8x + 4(x^2 - 2) \times -2e^{-2x}$	M1 A1	1.1b 1.1b
	$f'(x) = 8e^{-2x} \{x - (x^2 - 2)\} = 8(2 + x - x^2)e^{-2x}$ *	A1*	2.1
		(3)	
(b)	States roots of $f'(x) = 0$ $x = -1, 2$	B1	1.1b
	Substitutes one x value to find a y value	M1	1.1b
	Stationary points are $(-1, -4e^2)$ and $(2, 8e^{-4})$	A1	1.1b
		(3)	
(c)	(i) Range $[-8e^2, \infty)$ o.e. such as $g(x) \geq -8e^2$	B1ft	2.5
	(ii) For - Either attempting to find $2f(0) - 3 = 2 \times -8 - 3 = (-19)$ and identifying this as the lower bound - Or attempting to find $2 \times "8e^{-4}" - 3$ and identifying this as the upper bound	M1	3.1a
	Range $[-19, 16e^{-4} - 3]$	A1	1.1b
		(3)	
(9 marks)			
Notes:			

(a)

M1: Attempts the product rule and uses $e^{-2x} \rightarrow ke^{-2x}$, $k \neq 0$

If candidate states $u = 4(x^2 - 2)$, $v = e^{-2x}$ with $u' = \dots$, $v' = \dots e^{-2x}$ it can be implied by their $vu' + uv'$

If they just write down an answer without working award for $f'(x) = pxe^{-2x} \pm q(x^2 - 2)e^{-2x}$

They may multiply out first $f(x) = 4x^2e^{-2x} - 8e^{-2x}$. Apply in the same way condoning slips

Alternatively attempts the quotient rule on $f(x) = \frac{u}{v} = \frac{4(x^2 - 2)}{e^{2x}}$ with $v' = ke^{2x}$ and $f'(x) = \frac{vu' - uv'}{v^2}$

A1: A correct $f'(x)$ which may be unsimplified.

Via the quotient rule you can award for $f'(x) = \frac{8xe^{2x} - 8(x^2 - 2)e^{2x}}{e^{4x}}$ o.e.

A1*: Proceeds correctly to given answer showing all necessary steps.

The $f'(x)$ or $\frac{dy}{dx}$ must be present at some point in the solution

This is a "show that" question and there must not be any errors. All bracketing must be correct.

Allow a candidate to move from the **simplified** unfactorised answer of $f'(x) = 8xe^{-2x} - 8(x^2 - 2)e^{-2x}$

to the given answer in one step.

Do not allow it from an **unsimplified** $f'(x) = 4 \times 2xe^{-2x} + 4(x^2 - 2) \times -2e^{-2x}$

Allow the expression / bracketed expression to be written in a different order.

So, for example, $8(x - x^2 + 2)e^{-2x}$ is OK

(b)

B1: States or implies $x = -1, 2$ (as the roots of $f'(x) = 0$)

M1: Substitutes one x value of their solution to $f'(x) = 0$ in $f(x)$ to find a y value.

Allow decimals here (3sf). FYI, to 3 sf, $-4e^2 = -29.6$ and $8e^{-4} = 0.147$

Some candidates just write down the x coordinates but then go on in part (c) to find the ranges using the y coordinates. Allow this mark to be scored from work in part (c)

A1: Obtains $(-1, -4e^2)$ and $(2, 8e^{-4})$ as the stationary points. This must be scored in (b). Remember to isw

after a correct answer. Allow these to be written separately. E.g. $x = -1, y = -4e^2$

Extra solutions, e.g. from $x = 0$ will be penalised on this mark.

(c)(i)

B1ft: For a correct range written using correct notation.

Follow through on $2 \times$ their minimum "y" value from part (b), providing it is negative.

Condone a decimal answer if this is consistent with their answer in (b) to 3sf or better.

Examples of correct responses are $[-8e^2, \infty)$, $g \geq -8e^2$, $y \geq -8e^2$, $\{q \in \mathbb{R}, q \geq -8e^2\}$

(c)(ii)

M1: See main scheme. Follow through on $2 \times$ their " $8e^{-4}$ " $- 3$ for the upper bound.

A1: Range $[-19, 16e^{-4} - 3]$ o.e. such as $-19 \leq y \leq 16e^{-4} - 3$ but must be exact

Q11.

Question	Scheme	Marks	AOs
(a)	$x^2 \tan y = 9 \Rightarrow 2x \tan y + x^2 \sec^2 y \frac{dy}{dx} = 0$	M1 A1	3.1a 1.1b
	Full method to get $\frac{dy}{dx}$ in terms of x using $\sec^2 y = 1 + \tan^2 y = 1 + f(x)$	M1	1.1b
	$\frac{dy}{dx} = \frac{-2x \times \frac{9}{x^2}}{x^2 \left(1 + \frac{81}{x^4}\right)} = \frac{-18x}{x^4 + 81} \quad *$	A1*	2.1
		(4)	
(b)	$\frac{dy}{dx} = \frac{-18x}{x^4 + 81}$ $\Rightarrow \frac{d^2y}{dx^2} = \frac{-18 \times (x^4 + 81) - (-18x)(4x^3)}{(x^4 + 81)^2} = \frac{54(x^4 - 27)}{(x^4 + 81)^2} \text{ o.e.}$	M1 A1	1.1b 1.1b
	States that when $x < \sqrt[4]{27} \Rightarrow \frac{d^2y}{dx^2} < 0$ when $x = \sqrt[4]{27} \Rightarrow \frac{d^2y}{dx^2} = 0$ AND when $x > \sqrt[4]{27} \Rightarrow \frac{d^2y}{dx^2} > 0$ giving a point of inflection when $x = \sqrt[4]{27}$	A1	2.4
		(3)	
		(7 marks)	
Notes:			

(a)

M1: Attempts to differentiate $\tan y$ implicitly. Eg. $\tan y \rightarrow \sec^2 y \frac{dy}{dx}$ or $\cot y \rightarrow -\operatorname{cosec}^2 y \frac{dy}{dx}$

You may well see an attempt $\tan y = \frac{9}{x^2} \Rightarrow \sec^2 y \frac{dy}{dx} = \dots$

When a candidate writes $x^2 \tan y = 9 \Rightarrow x = 3 \tan^{\frac{1}{2}} y$ the mark is scored for $\tan^{\frac{1}{2}} y \rightarrow \dots \tan^{\frac{3}{2}} y \sec^2 y$

A1: Correct differentiation $2x \tan y + x^2 \sec^2 y \frac{dy}{dx} = 0$

Allow also $\sec^2 y \frac{dy}{dx} = -\frac{18}{x^3}$ or $2x = -9 \operatorname{cosec}^2 y \frac{dy}{dx}$ amongst others

M1: Full method to get $\frac{dy}{dx}$ in terms of x using $\sec^2 y = 1 + \tan^2 y = 1 + f(x)$

A1*: Proceeds correctly to the given answer of $\frac{dy}{dx} = \frac{-18x}{x^4 + 81}$

(b)

M1: Attempts to differentiate the given expression using the product or quotient rule.

For example look for a correct attempt at $\frac{vu' - uv'}{v^2}$ with $u = -18x, v = x^4 + 81, u' = \pm 18, v' = \dots x^3$

If no method is seen or implied award for $\frac{\pm 18 \times (x^4 + 81) \pm 18x(ax^3)}{(x^4 + 81)^2}$

Using the product rule award for $\pm 18(x^4 + 81)^{-1} \pm 18x(x^4 + 81)^{-2} \times cx^3$

A1: Correct **simplified** $\frac{d^2y}{dx^2} = \frac{54(x^4 - 27)}{(x^4 + 81)^2}$ o.e. such as $\frac{d^2y}{dx^2} = \frac{54x^4 - 1458}{(x^4 + 81)^2}$

Alternatively score for showing that when a correct (unsimplified) $\frac{d^2y}{dx^2} = 0 \Rightarrow x^4 = 27 \Rightarrow x = \sqrt[4]{27}$

Or for substituting $x = \sqrt[4]{27}$ into an unsimplified but correct $\frac{d^2y}{dx^2}$ and showing that it is 0

A1: Correct explanation with a minimal conclusion and correct second derivative.

See scheme.

It can be also be argued from $x^4 < 27, x^4 = 27$ and $x^4 > 27$ provided the conclusion states that the point of inflection is at $x = \sqrt[4]{27}$

Alternatively substitutes values of x either side of $\sqrt[4]{27}$ and at $\sqrt[4]{27}$, into $\frac{d^2y}{dx^2}$, finds all three values and makes a minimal conclusion.

A different method involves finding $\frac{d^3y}{dx^3}$ and showing that $\frac{d^3y}{dx^3} \neq 0$ and $\frac{d^2y}{dx^2} = 0$ when $x = \sqrt[4]{27}$

$$\text{FYI } \frac{d^3y}{dx^3} = \frac{23328x^3}{(x^4 + 81)^3} = 0.219 \text{ when } x = \sqrt[4]{27}$$

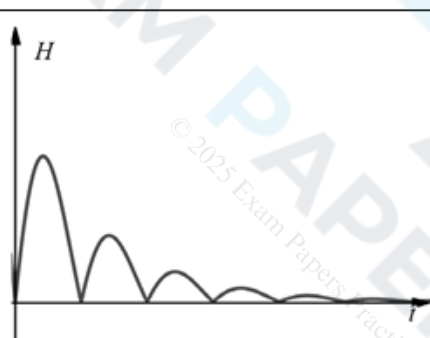
Alternative part (a) using arctan

M1: Sets $y = \arctan \frac{9}{x^2} \rightarrow \frac{dy}{dx} = \frac{1}{1 + \left(\frac{9}{x^2}\right)^2} \times \dots$ where $\dots$ could be 1

A2: $y = \arctan \frac{9}{x^2} \rightarrow \frac{dy}{dx} = \frac{1}{1 + \left(\frac{9}{x^2}\right)^2} \times -\frac{18}{x^3}$

A1*: $\frac{dy}{dx} = \frac{1}{1 + \frac{81}{x^4}} \times -\frac{18}{x^3} = \frac{-18x}{x^4 + 1}$ showing correct intermediate step and no errors.

Q12.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$f'(x) = -2.5e^{-0.25x} \sin x + 10e^{-0.25x} \cos x$	M1	This mark is given for a method to differentiate to find an expression for $f'(x)$
		A1	This mark is given for correctly differentiating to find an expression for $f'(x)$
	$f'(x) = 0$ $\Rightarrow e^{-0.25x} (-2.5 \sin x + 10 \cos x) = 0$ $\Rightarrow (-2.5 \sin x + 10 \cos x) = 0$	M1	This mark is given for setting $f'(x) = 0$ and finding a method to solve for $\tan x$
	$\frac{\sin x}{\cos x} = \frac{10}{2.5}$ $\tan x = 4$	A1	This mark is given for showing that $\tan x = 4$ as required.
(b)		M1	This mark is given for a graph with a correct shape
		A1	This mark is given for a graph with heights > 0
(c)	$\tan x = 4, x = 1.326$ $t = \pi + 1.326 = 4.47$	M1	This method is given for finding a value for t between the first and second bounce
	$H(4.47) = 10e^{-0.25 \times 4.47} \sin 4.47 $	M1	This mark is given for substituting the value of $t = \pi + \arctan 4$ into $H(t)$
	$= 3.27 \times -0.97 $ $= 3.17 \text{ metres}$	A1	This mark is given for finding the maximum height of the ball
(d)	The time between each bounce should not stay the same when the heights of each bounce are getting smaller	B1	This mark is given for a valid explanation of why the model should not be used to predict the time of each bounce

Q13.

Question	Scheme	Marks	AOs
(a)	Sets $500 = \pi r^2 h$	B1	2.1
	Substitute $h = \frac{500}{\pi r^2}$ into $S = 2\pi r^2 + 2\pi r h = 2\pi r^2 + 2\pi r \times \frac{500}{\pi r^2}$	M1	2.1
	Simplifies to reach given answer $S = 2\pi r^2 + \frac{1000}{r}$ *	A1*	1.1b
		(3)	
(b)	Differentiates S with both indices correct in $\frac{dS}{dr}$	M1	3.4
	$\frac{dS}{dr} = 4\pi r - \frac{1000}{r^2}$	A1	1.1b
	Sets $\frac{dS}{dr} = 0$ and proceeds to $r^3 = k$, k is a constant	M1	2.1
	Radius = 4.30 cm	A1	1.1b
	Substitutes their $r = 4.30$ into $h = \frac{500}{\pi r^2} \Rightarrow$ Height = 8.60 cm	A1	1.1b
		(5)	
(c)	States a valid reason such as - The radius is too big for the size of our hands - If $r = 4.3$ cm and $h = 8.6$ cm the can is square in profile. All drinks cans are taller than they are wide - The radius is too big for us to drink from - They have different dimensions to other drinks cans and would be difficult to stack on shelves with other drinks cans	B1	3.2a
		(1)	
9 marks			

Notes:
(a) B1: Uses the correct volume formula with $V=500$. Accept $500 = \pi r^2 h$ M1: Substitutes $h = \frac{500}{\pi r^2}$ or $rh = \frac{500}{\pi r}$ into $S = 2\pi r^2 + 2\pi r h$ to get S as a function of r A1*: $S = 2\pi r^2 + \frac{1000}{r}$ Note that this is a given answer.
(b) M1: Differentiates the given S to reach $\frac{dS}{dr} = Ar \pm Br^{-2}$ A1: $\frac{dS}{dr} = 4\pi r - \frac{1000}{r^2}$ or exact equivalent M1: Sets $\frac{dS}{dr} = 0$ and proceeds to $r^3 = k$, k is a constant A1: $R =$ awrt 4.30cm A1: $H =$ awrt 8.60 cm
(c) B1: Any valid reason. See scheme for alternatives

Q14.

Question	Scheme	Marks	AOs
(a)	(i) $\frac{dy}{dx} = 2x - 2 - 12x^{-\frac{1}{2}}$	M1 A1	1.1b 1.1b
	(ii) $\frac{d^2y}{dx^2} = 2 + 6x^{-\frac{3}{2}}$	B1ft	1.1b
		(3)	
(b)	Substitutes $x = 4$ into their $\frac{dy}{dx} = 2 \times 4 - 2 - 12 \times 4^{-\frac{1}{2}} = \dots$	M1	1.1b
	Shows $\frac{dy}{dx} = 0$ and states "hence there is a stationary point" oe	A1	2.1
		(2)	
(c)	Substitutes $x = 4$ into their $\frac{d^2y}{dx^2} = 2 + 6 \times 4^{-\frac{3}{2}} = (2.75)$	M1	1.1b
	$\frac{d^2y}{dx^2} = 2.75 > 0$ and states "hence minimum"	A1ft	2.2a
		(2)	
(7 marks)			

(a)(i)

M1: Differentiates to $\frac{dy}{dx} = Ax + B + Cx^{-\frac{1}{2}}$ A1: $\frac{dy}{dx} = 2x - 2 - 12x^{-\frac{1}{2}}$ (Coefficients may be unsimplified)

(a)(ii)

B1ft: Achieves a correct $\frac{d^2y}{dx^2}$ for their $\frac{dy}{dx}$ (Their $\frac{dy}{dx}$ must have a negative or fractional index)

(b)

M1: Substitutes $x = 4$ into their $\frac{dy}{dx}$ and attempts to evaluate. There must be evidence $\frac{dy}{dx}|_{x=4} = \dots$

Alternatively substitutes $x = 4$ into an equation resulting from $\frac{dy}{dx} = 0$ Eg. $\frac{36}{x} = (x-1)^2$ and equates

A1: There must be a reason and a minimal conclusion. Allow $\checkmark$, QED for a minimal conclusion

Shows $\frac{dy}{dx} = 0$ and states "hence there is a stationary point" oe

Alt Shows that $x = 4$ is a root of the resulting equation and states "hence there is a stationary point"

All aspects of the proof must be correct including a conclusion

(c)

M1: Substitutes $x = 4$ into their $\frac{d^2y}{dx^2}$ and calculates its value, or implies its sign by a statement such as

when $x = 4 \Rightarrow \frac{d^2y}{dx^2} > 0$. This must be seen in (c) and not labelled (b). Alternatively calculates the

gradient of C either side of $x = 4$ or calculates the value of y either side of $x = 4$.

A1ft: For a correct calculation, a valid reason and a correct conclusion. Ignore additional work where

candidate finds $\frac{d^2y}{dx^2}$ left and right of $x = 4$. Follow through on an incorrect $\frac{d^2y}{dx^2}$ but it is dependent upon

having a negative or fractional index. Ignore any references to the word convex. The nature of the turning point is "minimum".

Using the gradient look for correct calculations, a valid reason.... goes from negative to positive, and a correct conclusion ...minimum.

Q15.

Question	Scheme	Marks	AOs
(a)	Either $3y^2 \rightarrow Ay \frac{dy}{dx}$ or $2xy \rightarrow 2x \frac{dy}{dx} + 2y$	M1	2.1
	$2x - 2x \frac{dy}{dx} - 2y + 6y \frac{dy}{dx} = 0$	A1	1.1b
	$(6y - 2x) \frac{dy}{dx} = 2y - 2x$	M1	2.1
	$\frac{dy}{dx} = \frac{2y - 2x}{6y - 2x} = \frac{y - x}{3y - x} *$	A1*	1.1b
	(4)		
(b)	$\left(\text{At } P \text{ and } Q \frac{dy}{dx} \rightarrow \infty \Rightarrow \right)$ Deduces that $3y - x = 0$	M1	2.2a
	Solves $y = \frac{1}{3}x$ and $x^2 - 2xy + 3y^2 = 50$ simultaneously	M1	3.1a
	$\Rightarrow x = (\pm)5\sqrt{3}$ OR $\Rightarrow y = (\pm)\frac{5}{3}\sqrt{3}$	A1	1.1b
	Using $y = \frac{1}{3}x \Rightarrow x = ..$ AND $y = ..$	dM1	1.1b
	$P = \left(-5\sqrt{3}, -\frac{5}{3}\sqrt{3} \right)$	A1	2.2a
	(5)		
(c)	Explains that you need to solve $y = x$ and $x^2 - 2xy + 3y^2 = 50$ simultaneously and choose the positive solution	B1ft	2.4
	(1)		
(10 marks)			

Notes:

(a)

M1: For selecting the appropriate method of differentiating either $3y^2 \rightarrow Ay \frac{dy}{dx}$ or $2xy \rightarrow 2x \frac{dy}{dx} + 2y$

It may be quite difficult awarding it for the product rule but condone $-2xy \rightarrow -2x \frac{dy}{dx} + 2y$ unless you see evidence that they have used the incorrect law $vu' - uv'$

A1: Fully correct derivative $2x - 2x \frac{dy}{dx} - 2y + 6y \frac{dy}{dx} = 0$

Allow attempts where candidates write $2xdx - 2xdy - 2ydx + 6ydy = 0$

but watch for students who write $\frac{dy}{dx} = 2x - 2x \frac{dy}{dx} - 2y + 6y \frac{dy}{dx}$ This, on its own, is A0 unless you are

convinced that this is just their notation. Eg $\frac{dy}{dx} = 2x - 2x \frac{dy}{dx} - 2y + 6y \frac{dy}{dx} = 0$

M1: For a valid attempt at making $\frac{dy}{dx}$ the subject, with two terms in $\frac{dy}{dx}$ coming from $3y^2$ and $2xy$

Look for $(\dots \pm \dots) \frac{dy}{dx} = \dots$ It is implied by $\frac{dy}{dx} = \frac{2y-2x}{6y-2x}$

This cannot be scored from attempts such as $\frac{dy}{dx} = 2x - 2x \frac{dy}{dx} - 2y + 6y$ which only has one correct term.

A1*: $\frac{dy}{dx} = \frac{y-x}{3y-x}$ with no errors or omissions.

The previous line $\frac{dy}{dx} = \frac{2y-2x}{6y-2x}$ or equivalent must be seen.

(b)

M1: Deduces that $3y - x = 0$ oe

M1: Attempts to find either the x or y coordinates of P and Q by solving their $y = \frac{1}{3}x$ with

$x^2 - 2xy + 3y^2 = 50$ simultaneously. Allow for finding a quadratic equation in x or y and solving to find at least one value for x or y .

This may be awarded when candidates make the numerator = 0 ie using $y = x$

A1: $\Rightarrow x = (\pm)5\sqrt{3}$ OR $\Rightarrow y = (\pm)\frac{5}{3}\sqrt{3}$

dM1: Dependent upon the previous M, it is for finding the y coordinate from their x (or vice versa)

This may also be scored following the numerator being set to 0 ie using $y = x$

A1: Deduces that $P = \left(-5\sqrt{3}, -\frac{5}{3}\sqrt{3}\right)$ OE. Allow to be $x = \dots$ $y = \dots$

(c)

B1ft: Explains that this is where $\frac{dy}{dx} = 0$ and so you need to solve $y = x$ and $x^2 - 2xy + 3y^2 = 50$

simultaneously and choose the positive solution (or larger solution).

Allow a follow through for candidates who mix up parts (b) and (c)

Alternatively candidates could complete the square $(x-y)^2 + 2y^2 = 50$ and state that y would reach a maximum value when $x = y$ and choose the positive solution from $2y^2 = 50$